AOFC Exercise 05 18.11.2025

Problem 9

a) $n_1 = 1.457$ $n_2 = 1.444$ $a = 6.1\,\mu m$ $\lambda = 1.55\,\mu m$

$$V = k_0 \cdot a \sqrt{n_1^2 - n_2^2} = \frac{2\pi}{1.55\,\mu m} \cdot 6.1\,\mu m \cdot \sqrt{1.457^2 - 1.444^2}$$

$$= 4.807$$

$B_{01} = 0.83$ $B_{11} = 0.58$; $B_{21} = 0.26$; $B_{02} = 0.17$

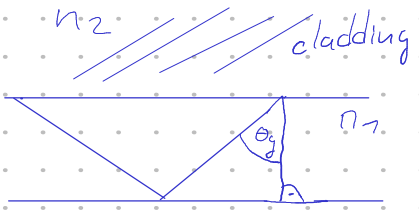
$$-\frac{v \cdot J_{\ell+1}(v)}{J_\ell} + \frac{v \cdot K_{\ell+1}(v)}{K_\ell(v)} = 0 \quad \text{with } V^2 = v^2 + w^2$$

$$-\frac{v \cdot J_{\ell+1}(v)}{J_\ell} + \frac{\sqrt{V^2 - v^2}\, K_{\ell+1}\left(\sqrt{V^2 - v^2}\right)}{K_\ell\left(\sqrt{V^2 - v^2}\right)} = f(v)$$

$B_{01} = 0.83$ $B_{11} = 0.58$; $B_{21} = 0.26$ $B_{02} = 0.17$

$\beta = k_0 \cdot n_1 \cdot \sin\theta_g$ $B = \frac{\frac{\beta}{k_0} - n_2}{n_1 - n_2}$ $\frac{\beta}{k_0} = B \cdot (n_1 - n_2) + n_2$

$$\frac{\beta}{k_0} = n_1 \cdot \sin\theta_g$$



$$\theta_g = \arcsin\left(\frac{B \cdot (n_1 - n_2) + n_2}{n_1}\right) = 86.84° \rightarrow LP_{01}$$

$= 85.04° \rightarrow LP_{11}$

$= 83.41°$ LP_{21}

$= 83.02°$ LP_{02}

b) we are now looking at the group delay of the different modes inside the fiber

How is τ defined $\tau = \frac{d\beta}{d\omega}$ & $\beta = B \cdot k_0 (n_1 - n_2)$

$$\tau = \frac{d\left(B \cdot k_0 (n_1 - n_2)\right)}{d\omega} \qquad \frac{dn_1}{d\omega} = \frac{dn_2}{d\omega}$$

$$\tau = \underbrace{\frac{(n_1 - n_2)}{c} \cdot V \frac{d(Vb)}{dV}}_{\text{waveguide delay}} + \underbrace{\frac{1}{c} \cdot N_2}_{\text{delay material}}$$

Same modes

$V = 4.802$

$$V\frac{d(Vb)}{dV}\Big|_{LP_{01}} = 1.11 \quad ; \quad \Big|_{LP_{11}} = 1.245 \quad \Big|_{LP_{21}} = 1.36 \quad \Big|_{LP_{02}} = 1.27$$

$$\tau_{01} = \frac{n_1 - n_2}{c} \cdot V\frac{d(Vb)}{dV} = \frac{1.457 - 1.444}{3e8\ \frac{m}{s}} \cdot 1.11 = 48.1\ \frac{ps}{m}$$ fastest mode

τ_{11} $1.245 = 53.95\ ps/m$

τ_{21} $1.36 = 58.93\ ps/m$ slowest mode

τ_{02} $1.27 = 55.03\ ps/m$

c) Length: 2,3 km

$$t_{01} = 2{,}3e3\ m \cdot 48.1\ \frac{ps}{m} = 110.63\ us$$
$$t_{21} = 2{,}3e3\ m \cdot 58{,}93\ \frac{ps}{m} = 135{,}54\ us$$
$\Delta t \approx 25ns$ → differential mode delay (DMD)

Measurement of the DMD: one posibility, launching pulse into a mode-multiplexer, and check the routine of different modes.

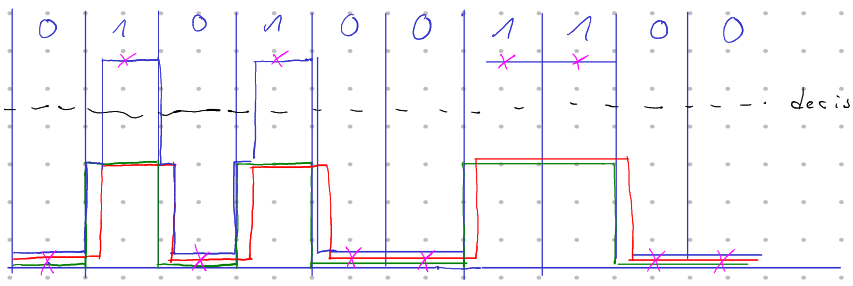
OR usage of an Optical Vector Network analyzer (OVNA)

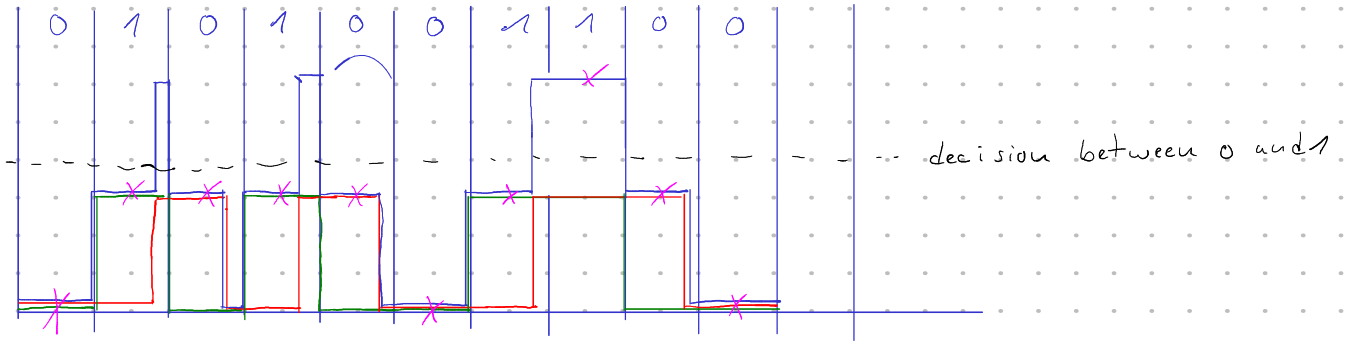
d) first we normalize the graphs to the corresponding error rates

$$\tilde{\Delta t} = \Delta t \cdot R_s = \frac{\Delta t}{T_s}$$

10 MBd $\Delta t_{10} = 25ns \cdot 10\ MBd = 0.25$ Symbols differential Mode delay (→ 10e6 Symbols/s)

30 MBd $\Delta t_{30} = 25ns \cdot 30e6\ Symbols/s = 0.75$ Symbols





Sn 0 1 0 1 0 0 1 1 0 0

10MBd 0 1 0 1 0 0 1 1 0 0 → 0% BER

30MBd 0 0 0 0 0 0 0 1 0 0 → 30% BER

$$\text{bit error rate} = \frac{\text{incorrect received bits}}{\text{transmitted bits}}$$

Problem 10

a) Dispersion coefficient

$$\tau = \frac{n_1 - n_2}{c} \cdot V \cdot \frac{d(Vb)}{dV} + \frac{1}{c} N_2 \quad \text{we need the derivative } \tau \text{ over } \lambda$$

$$\frac{d\tau}{d\lambda} = -\frac{n_1 - n_2}{c \cdot \lambda} \cdot V \cdot \frac{d^2(Vb)}{dV^2} - \frac{\lambda}{c} \cdot \frac{d^2 n_2}{d\lambda}$$

comes out of Diagramm

waveguide dispersion D_W

material dispersion, known before.

$$\| \quad 17 \frac{ps}{nm\,km} = D_M$$

$a = 4\,\mu m \qquad NA = 0.14 \qquad n_2 = 1.44 \qquad \lambda = 1.55\,\mu m$

$$V = k_0 \cdot a \cdot A_N = \frac{2\pi}{1.55\mu m} \cdot 4\mu m \cdot 0.14 = 2.27$$

$$d\,\frac{d^2(Vb)}{dV^2} = 0.27 \; !$$

$$NA^2 = n_1^2 - n_2^2$$

$$n_1 = \sqrt{NA^2 + n_2^2} = 1.447$$

$$D_W = -\frac{n_1 - n_2}{c \cdot \lambda} \cdot V \frac{d^2(Vb)}{dV^2} = -\frac{1.447 - 1.44}{3e8 \cdot 1.55\mu m} \cdot 0.27 = -4.065 \frac{ps}{nm\,km}$$

$$D = D_W + D_M = -4.065 \frac{ps}{nm\,km} + 17 \frac{ps}{nm\,km} = 12.935 \frac{ps}{nm\,km}$$ → Dispersion of the LP_{01} mode!

→ other modes will have different dispersion!

$\frac{S}{\omega}$